

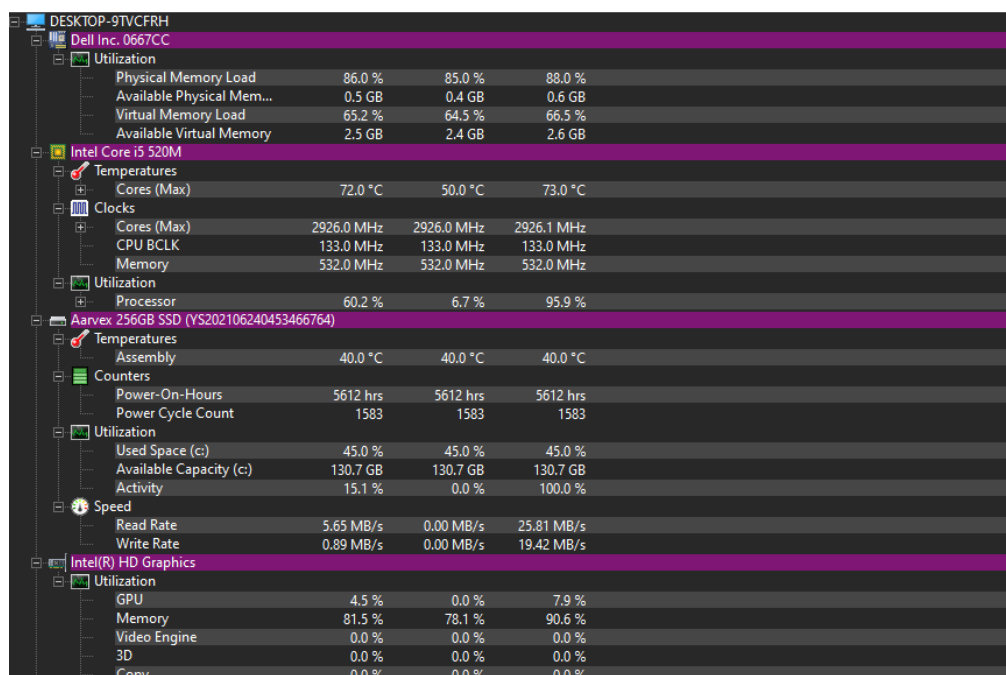
1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

- **Computer won't power on** – A multimeter measures the power supply output voltages to verify whether the PSU is delivering the correct power to the motherboard.
- **Smartphone won't charge** – A multimeter checks the charger, USB cable, and battery voltage to identify whether the charging components are functioning properly.
- **Laptop or smartphone has a suspected short circuit** – A multimeter uses continuity or resistance mode to detect shorted components or broken connections on the circuit board.

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video

- The POST card displayed hexadecimal POST codes while the motherboard performed its startup tests.
- Each code showed the current stage of the boot process and where it stopped. In the video, the POST card identified a hardware problem by stopping at a specific POST code.
- This helped determine that the issue was with the RAM, making troubleshooting faster.

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.



4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it Checks	Example Problem Detected
Multimeter	Measures voltage, current, resistance, and continuity in electrical components.	Detects a faulty power supply that is not providing the correct voltage.
POST Card	Displays motherboard POST (Power-On Self-Test) error codes during startup.	Identifies faulty RAM or a motherboard hardware error preventing the computer from booting.
Software Diagnostics	Checks system hardware health, temperatures, storage status, memory, and performance.	Detects an overheating CPU, failing hard drive/SSD, or high memory usage.